Section 2.4
Partitioned Matrices

Elementary Linear Algebra

(MATH 2250-03)

Spring, 2018

A key feature of our work with matrices has been the ability to regard a matrix A as a list of column vectors rather than just a rectangular array of numbers. This point of view has been so useful that we wish to consider other partitions of A, indicated by horizontal and vertical dividing rules.

Example 1. The matrix

$$A = \left[\begin{array}{ccc|cc|c} 1 & 2 & 3 & 4 & 5 & 6 \\ 7 & 8 & 9 & 1 & 2 & 3 \\ \hline 4 & 5 & 6 & 7 & 8 & 9 \end{array}\right]$$

can also be written as the 2×3 *partitioned* (or *block*) *matrix*

$$A = \begin{bmatrix} A_{11} & A_{12} & A_{13} \\ A_{21} & A_{22} & A_{23} \end{bmatrix}$$

whose entries are the *blocks* or *submatrices*

$$A_{11} = \begin{bmatrix} 1 & 2 & 3 \\ 7 & 8 & 9 \end{bmatrix} \quad A_{12} = \begin{bmatrix} 4 & 5 \\ 1 & 2 \end{bmatrix} \quad A_{13} = \begin{bmatrix} 6 \\ 3 \end{bmatrix}$$

$$A_{21} = \begin{bmatrix} 4 & 5 & 6 \end{bmatrix} \quad A_{22} = \begin{bmatrix} 7 & 8 \end{bmatrix} \quad A_{23} = \begin{bmatrix} 9 \end{bmatrix}$$

Addition and Scalar Multiplication

If matrices A and B are the same size and are partitioned in exactly the same way, then it is natural to make the same partition of the ordinary matrix sum $A + B$. In this case, each block of $A + B$ is the (matrix) sum of the corresponding blocks of A and B. Multiplication of a partitioned matrix by a scalar is also computed block by block.

Multiplication of Partitioned Matrices

Partitioned matrices can be multiplied by the usual row-column rule as if the block entries were scalars, provided that for a product AB, the column partition of A matches the row partition of B.

Example 2. Let

$$A = \left[\begin{array}{ccc|cc} 2 & -3 & 1 & 0 & -4 \\ 1 & 5 & -2 & 3 & -1 \\ \hline 0 & -4 & -2 & 7 & -1 \end{array}\right] = \begin{bmatrix} A_{11} & A_{12} \\ A_{21} & A_{22} \end{bmatrix}, \quad B = \left[\begin{array}{cc} 6 & 4 \\ -2 & 1 \\ -3 & 7 \\ \hline -1 & 3 \\ 5 & 2 \end{array}\right] = \begin{bmatrix} B_1 \\ B_2 \end{bmatrix}.$$

The 5 columns of A are partitioned into a set of 3 columns and then a set of 2 columns. The 5 rows of B are partitioned in the same way–into a set of 3 rows and then a set of 2 rows. We say that the partitions of A and B are *conformable* for block multiplication. It can be shown that the ordinary product AB can be written as

$$\mathrm{AB} = \begin{bmatrix} \mathrm{A}_{11} & \mathrm{A}_{12} \\ \mathrm{A}_{21} & \mathrm{A}_{22} \end{bmatrix} \begin{bmatrix} \mathrm{B}_1 \\ \mathrm{B}_2 \end{bmatrix} = \begin{bmatrix} \mathrm{A}_{11}\mathrm{B}_1 + \mathrm{A}_{12} + \mathrm{B}_2 \\ \mathrm{A}_{21}\mathrm{B}_1 + \mathrm{A}_{22}\mathrm{B}_2 \end{bmatrix} = \left[\begin{array}{cc} -5 & 4 \\ -6 & 2 \\ \hline 2 & 1 \end{array}\right]$$

It is important for each smaller product in the expression for AB to be written **with the submatrix from** A **on the left**, since matrix multiplication is not commutative. For instance,

$$\mathrm{A}_{11}\mathrm{B}_1 = \begin{bmatrix} 2 & -3 & 1 \\ 1 & 5 & -2 \end{bmatrix} \begin{bmatrix} 6 & 4 \\ -2 & 1 \\ -3 & 7 \end{bmatrix} = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix}$$

$$\mathrm{A}_{12}\mathrm{B}_2 = \begin{bmatrix} 0 & -4 \\ 3 & -1 \end{bmatrix} \begin{bmatrix} -1 & 3 \\ 5 & 2 \end{bmatrix} = \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix}$$

Hence the top block in AB is

$$\mathrm{A}_{11}\mathrm{B}_1 + \mathrm{A}_{12}\mathrm{B}_2 = \begin{bmatrix} 15 & 12 \\ 2 & -5 \end{bmatrix} + \begin{bmatrix} -20 & -8 \\ -8 & 7 \end{bmatrix} = \begin{bmatrix} -5 & 4 \\ -6 & 2 \end{bmatrix}$$

The row-column rule for multiplication of block matrices provides the most general way to regard the product of two matrices. Each of the following views of a product has already been described using simple partitions of matrices:

1. the definition of Ax using the columns of A,
2. the column definition of AB,
3. the row-column rule for computing AB , and
4. the rows of AB as products of the rows of A and the matrix B

A fifth view of AB, again using partitions, follows in the next Theorem.

Outer Products

In the following theorem, let $col_k(\mathrm{A})$ denote the k^{th} column of A and let $row_k(\mathrm{B})$ denote the k^{th} row of B.

Theorem 1. *If* A *is an* $m \times n$ *matrix and* B *is an* $n \times p$ *matrix, then*

$$\begin{aligned} \mathrm{AB} &= \begin{bmatrix} col_1(\mathrm{A}) & col_2(\mathrm{A}) & \ldots & col_n(\mathrm{A}) \end{bmatrix} \begin{bmatrix} row_1(\mathrm{B}) \\ row_2(\mathrm{B}) \\ \vdots \\ row_n(\mathrm{B}) \end{bmatrix} \\ &= col_1(\mathrm{A})row_1(\mathrm{B}) + col_2(\mathrm{A})row_2(\mathrm{B}) + \ldots + col_n(\mathrm{A}) + row_n(\mathrm{A}) \end{aligned}$$

Recalling that the outer product of two vectors u and v belonging to $\mathbb{R}^n$ is given by $u \otimes v = uv^{\mathrm{T}}$, observe that the above then reads

$$\mathrm{AB} = col_1(\mathrm{A}) \otimes \left(row_1(\mathrm{B})\right)^{\mathrm{T}} + col_2(\mathrm{A}) \otimes \left(row_2(\mathrm{B})\right)^{\mathrm{T}} + \ldots + col_n(\mathrm{A}) \otimes \left(row_n(\mathrm{B})\right)^{\mathrm{T}}$$

More compactly, we have

$$\mathrm{AB} = \sum_{i=1}^{n} col_i(\mathrm{A}) \otimes \left(row_i(\mathrm{B})\right)^{\mathrm{T}}$$